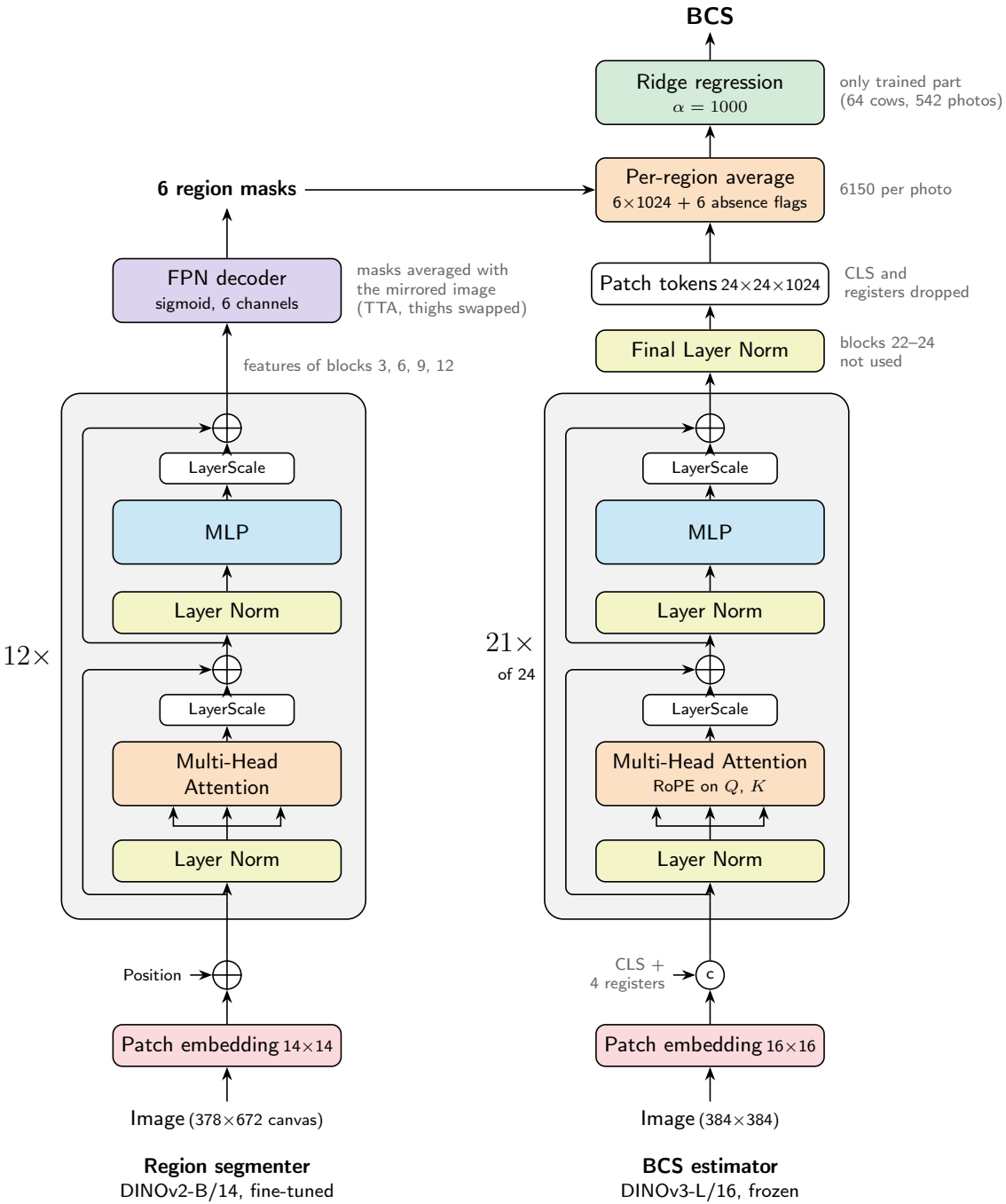

BCS
Ridge regression
$\alpha = 1000$
only trained part
(64 cows, 542 photos)
Per-region average
6×1024 + 6 absence flags
6150 per photo
6 region masks
FPN decoder
sigmoid, 6 channels
masks averaged with
the mirrored image
(TTA, thighs swapped)
Patch tokens 24×24×1024
CLS and
registers dropped
Final Layer Norm
blocks 22–24
not used
features of blocks 3, 6, 9, 12
LayerScale
MLP
Layer Norm
12×
LayerScale
Multi-Head
Attention
Layer Norm
21×
of 24
LayerScale
MLP
Layer Norm
LayerScale
Multi-Head Attention
RoPE on Q, K
Layer Norm
Position
CLS +
4 registers
C
Patch embedding 14×14
Patch embedding 16×16
Image (378×672 canvas)
Image (384×384)
Region segmenter
DINOv2-B/14, fine-tuned
40 annotated + 345 pseudo-labeled frames
BCS estimator
DINOv3-L/16, frozen